# Part 2 – Scenario 11: Station-Name Suggestions

## Scenario Explanation

The system stores many railway station names and must quickly suggest up to five names beginning with a given prefix. Station names may also be inserted or removed. The main requirement is that prefix operations should depend mainly on prefix length rather than scanning all stored names.

## Chosen Technique and Justification

A Trie (prefix tree) is appropriate because each character is represented by a tree level. Insertion and exact search take O(L), where L is the station-name length, while locating a prefix takes O(P), where P is the prefix length. Suggestions are obtained by visiting child links from A to Z, giving alphabetical order without scanning every station. Rejected alternative: a hash table is excellent for exact search but does not naturally support starts-with queries without extra indexing or scanning.

## Logic

Create a root with 26 child positions. During insertion, ignore spaces for navigation, create missing character nodes, and store the original station name at the terminal node. Exact search follows the normalized characters and checks the terminal flag. Prefix search follows the prefix, then recursively visits children from A through Z and records at most five terminal station names. Deletion unmarks the terminal node and removes nodes that are no longer needed.

## Complexity Analysis

| ***Operation*** | ***Complexity*** | ***Reason*** |
|---|---|---|
| Insert | O(L) | One node/path step per character. |
| Exact search | O(L) | Follow the station-name characters. |
| Delete | O(L) | Follow the path and clean unused nodes. |
| Find prefix | O(P) | Only the P prefix characters are followed. |
| Suggestions | O(P + output) | Find prefix, then traverse until 5 results. |
| Space | O(total stored characters) | Trie nodes represent stored characters. |

## Observation

The Trie produced the required suggestions in alphabetical order and returned at most five results for JA. Exact searches correctly identified existing and missing stations. Deleting JAGDALPUR made it unavailable, and reinserting it restored the result. The key observation is that locating a prefix follows only its characters, so the prefix-location cost depends on P rather than directly on the number of stations.

## Conclusion

The Trie is well suited to this workload because it directly supports prefix queries, dynamic insertion and deletion, exact search, and alphabetical suggestions. Its main trade-off is memory usage from the 26-child representation, but it provides predictable string-length-based operations.